

Chapter 47. Sinus Infections in the Ventilated Patient

Jean-Jacques Rouby; Qin Lu

Sinus Infections in the Ventilated Patient: Introduction

Infection of paranasal sinus cavities is a well-recognized cause of fever in mechanically ventilated patients.¹⁻⁷ Easy to detect in maxillary sinuses, the infectious process also frequently involves ethmoid, frontal, and sphenoid sinuses⁸ where the diagnosis is more difficult to establish.⁹ Infectious sinusitis represents an important reservoir of bacteria¹⁰ that may disseminate into the respiratory tract¹¹ and intracranially.¹² In contrast to community-acquired sinusitis, ventilator-associated sinusitis is often clinically silent in sedated critically ill patients and may be underdiagnosed if not systematically screened for in the presence of fever of unknown origin. In the absence of diagnosis and appropriate treatment, bacteremia,^{2,5} ventilator-associated pneumonia,¹³ and life-threatening complications, such as orbital infection,¹⁴ meningitis, mastoiditis, cerebral abscess, or thrombosis of the sinus cavernosus, may result.¹² Early detection and treatment of infectious maxillary sinusitis significantly reduces the incidence of ventilator-associated pneumonia and may decrease intensive care mortality.¹¹

Physiology of Paranasal Sinuses

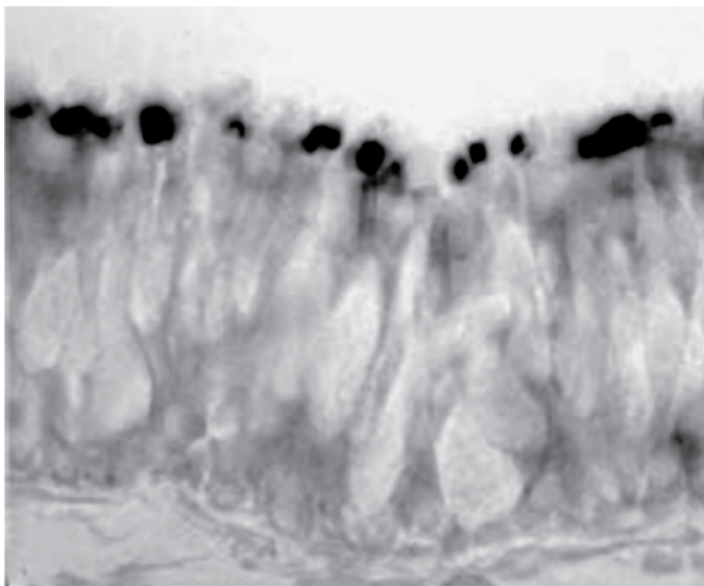
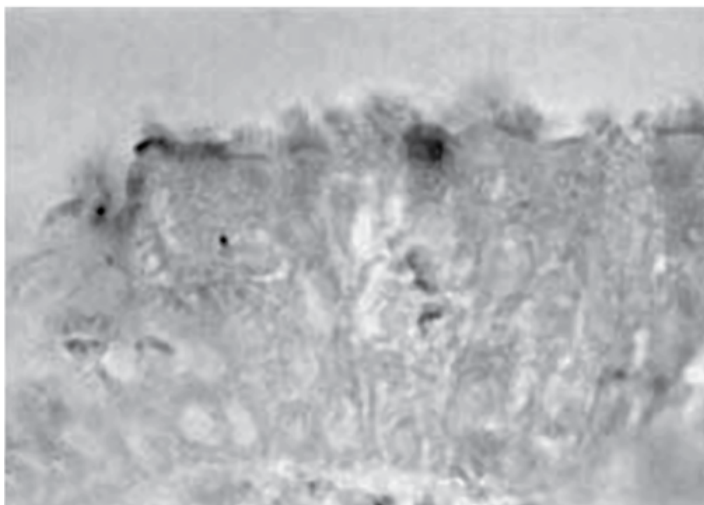
Classical Hypotheses

Located at the entry of the respiratory system, paranasal sinuses and the nose serve to humidify, filter, warm, and sense inspiratory gas.^{15,16} A number of other presupposed physiologic roles have not yet received firm scientific confirmation.¹⁷ Among them, the most popular are that paranasal sinuses help lighten the bones of the skull, improve the resonance of the voice, serve as a sound box for opera singers, increase the surface area of the olfactory membrane, serve as shock absorbers in mechanical impacts, protect against high pressure in the nasal region when sneezing, act as thermal insulators of the brain, and promote facial growth and architecture. More simply, paranasal sinuses might be considered as evolutionary relics or faults, whose form results from the influence of the forces created during the act of chewing.

Recent Hypothesis: The Role of the Regional Production of Nitric Oxide

Large amounts of **nitric oxide** are produced in human paranasal sinuses^{18,19} and permanently released in the upper airways through the different ostia that link antral cavities to the nostrils.²⁰ As shown in **Figure 47-1**, an inducible form of **nitric oxide** synthetase is present in cilia and microvilli of the maxillary sinus epithelium of healthy volunteers.^{18,19} Significant **nitric oxide** concentrations are also found in the exhaled gas of guinea pigs, pigs, rhesus monkeys, rats, rabbits, horses, and Asian elephants, all species possessing open paranasal sinuses.²⁰⁻²⁴ Interestingly, seals and baboons, which do not have open paranasal sinuses, do not exhale nitric oxide,^{25,26} thus confirming that pneumatic facial cavities play a critical role in the nasal production of **nitric oxide** in mammals. Humming, by accelerating gas exchange in sinus cavities, markedly increases exhaled nasal nitric oxide²⁷ whereas moderate exercise has an opposite effect.²⁸ **Nitric oxide** produced in the paranasal sinuses likely has an important role in host defense against inhaled pathogens, in the optimization of ventilation-perfusion ratios during normal breathing,²⁹ and in the regulation of ciliary motility.^{30,31}

Figure 47-1

**A****B**

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Immunoreactive sinus inducible **nitric oxide** synthetase distribution is represented in black on biopsy samples from maxillary sinus epithelium. In a healthy subject (**A**), a strong immunoreactivity is detected in the apical part of the ciliated epithelial cells near the surface. In a patient with radiographic maxillary sinusitis (**B**), the signal is weaker or absent. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Deja M, Busch T, Bachmann S, et al. Reduced **nitric oxide** in sinus epithelium of patients with radiologic maxillary sinusitis and sepsis. *Am J Respir Crit Care Med*. 2003;168:281–286. Official Journal of the American Thoracic Society.)

Each paranasal sinus communicates with the corresponding nostril by an ostium, which provides aeration of the antral cavity. Bacteria are present in nasal secretions covering the mucociliary epithelium and on the gingival mucosa and its crevices. Microorganisms may enter the maxillary cavity by the canine fossa and the inferior meatus. Paranasal sinuses, however, are normally sterile³² as a result of two different protective mechanisms. First, the antral mucosa is covered with a protective mucus layer produced by goblet cells. Bacteria penetrating into the maxillary cavity are immediately enveloped by mucus and moved rapidly to the sinus ostium by respiratory cilia. Second, the antral concentrations of **nitric oxide** exert a bacteriostatic effect. Many pathogens are sensitive to **nitric oxide** in concentrations less than 1 part per million.^{33–36} The antral concentrations of **nitric oxide**, produced by the maxillary epithelium, are equal to or greater than 2 parts per million and likely contribute to antral sterility.³²

Pathogenesis and Predisposing Factors

Pathogenesis of radiographic sinusitis diagnosed in intubated and tracheostomized patients is incompletely understood.

Experimental Models

Rabbits have large and accessible nostrils and their sinuses show anatomical similarities to those of humans. In this species, sinuses consist of a number of cavities connected to each other and can be considered as one, called the maxillary sinus.³⁷ Historically, the first experimental model was created by opening the maxillary sinus through an anterior wall antrostomy, gluing and closing the natural ostium, and injecting bacterial strains directly suppress “injected” into the maxillary cavity.³⁸ Although the model is extremely reliable for producing purulent sinusitis, it causes a traumatic injury that questions its relevance for human sinusitis. As a consequence, a rhinogenic model was proposed in the late 1990s and remains to date the reference:^{39,40} a foreign body—an endotracheal tube,⁴¹ a catheter,⁴² or a sponge impregnated or not with bacteria⁴³—is implanted into nostrils for a period of time ranging from several days to several weeks. In the rhinogenic rabbit model, computed tomographic evidence of maxillary sinusitis is detected at the sixth day following foreign-body nasal placement.⁴³

A number of lessons can be drawn from experimental studies. Direct or indirect obstruction of the ostia rapidly induces an acute sinus inflammation. The intranasal placement of a foreign body over several days results in purulent sinusitis, characterized by the accumulation of a thick and purulent discharge within the sinus cavity.⁴³ Initial histologic lesions are made of inflammation and exudation of maxillary mucosa; the number of goblet cells increases, augmenting mucus production and causing fluid accumulation within the cavity. Then, inflammatory polyp formation is observed,^{41,43,44} associated with permanent loss of cilia, squamous metaplasia, and fibrosis.^{43,45} The risk of sinusitis increases with the nasal catheterization period and the size of the intranasal catheter.^{41,42} Relief of ostial obstruction promotes resolution of radiographic sinusitis within a few days,⁴⁶ whereas sinus inflammation persists for several weeks.⁴³ Over time, the fluid-filled maxillary cavity is contaminated⁴¹ or infected⁴³ by species present in the adjacent nostril.⁴¹ Surprisingly, bacteria impregnating the intranasal sponge are rarely found in the nasal cavity and are often replaced by bacteria belonging to the normal flora of the rabbit nostrils.^{43,47}

Finally, the following sequence may lead to bacterial sinusitis. First, the presence of a nasal foreign body induces an ipsilateral inflammatory antral disease. Second, mucus accumulates within the antral cavity because of goblet cell proliferation, impairment of mucociliary clearance, and ostial obstruction. Third, the antral contents are contaminated and infected by microorganisms issued from the nostrils whose proliferation into the sinus cavity is facilitated by the lack of regional nitric oxide production.

Macroscopic and Microscopic Aspects of Sinusitis Complicating Mechanical Ventilation

Only a few studies have reported macroscopic and histologic disorders characterizing radiographic maxillary sinusitis in critically ill patients.^{19,48–50} In thirty-three neurosurgical patients receiving prolonged mechanical ventilation with radiographic maxillary sinusitis, antroscopies were performed by the canine fossa route.⁴⁹ Most of radiographic maxillary sinusitis corresponded to “inflammatory” sinusitis, with the antral cavity filled with thin and transparent secretions and/or gelatinous and transparent mucus with a neutral smell. In a minority of patients, bacterial sinusitis was diagnosed as the presence of viscous and opaque pus with a foul smell. Both types of sinusitis were associated with varying mucosal reactions, ranging from pallid to vermilion edema with different degrees of transparency. Interestingly, in two patients with “inflammatory” sinusitis and one patient with bacterial sinusitis, maxillary mucosa was macroscopically normal. When present, mucosal edema always included the medial side with the ostial region.⁴⁹ Biopsy specimens of antral mucosa revealed various grades of inflammation. Acute inflammation involving the maxillary epithelium, the connective tissue, and the vessels with a massive infiltration by polymorphonuclear neutrophils was observed in patients with bacterial maxillary sinusitis. In patients with “inflammatory” sinusitis, mild inflammation was observed, characterized by a moderate cellular infiltration of the connective tissue with a massive infiltration of vessels by eosinophils.⁴⁸

Predisposing Factors

Foreign Bodies Within the Nostrils

Several prospective and retrospective studies have shown that the occurrence of radiographic sinusitis has a higher incidence in the antrum adjacent to nasotracheal and nasogastric tubes than in the nonadjacent sinus.^{3,4,6,7,51–62} The presence of a nasogastric tube for enteral feeding appears as an independent risk factor.⁶³ As in experimental sinusitis, the presence of a foreign body within the nostrils creates an ostial obstruction that initiates sinus inflammation and mucus accumulation within the sinus cavity. Confirming experimental data,⁴² a large endotracheal tube induces radiographic sinusitis more frequently and faster than a small nasogastric tube.^{55,58} In addition, the plastic nasotracheal tube is the site of biofilm formation for bacteria with adhesive capacity and enhanced pathogenicity:^{64–66} microorganisms adhere to the internal and external surfaces of the endotracheal tube and some species exude an exopolysaccharide that acts as a slimelike adhesive.⁶⁷ Bacteria encased in this biofilm become partially resistant to the action of antimicrobials and host defences.^{68,69} The bacterial proliferation around nasal foreign bodies forms a reservoir from which microorganisms penetrate into the antral cavity.⁴¹

Size of the Meatus Inferior and Body Position

Ostial size, upon which drainage and ventilation of the sinus cavities depends, can markedly vary from one patient to another.⁷⁰ In humans, supine position is known to reduce ostial patency⁷¹ by inducing swelling of the ostiomeatal complex.⁷² A functional ostial area less than 5 mm² induces hypoventilation of the maxillary cavity, local hypoxia, decrease in mucosal blood flow, and predispose to maxillary sinusitis by impairing mucociliary clearance.⁷³

Sepsis

Generalized sepsis, well known for stimulating numerous tissue-inducible **nitric oxide** synthetases, inhibits the **nitric oxide** metabolic pathway at the maxillary level.¹⁹ Autoinhaled **nitric oxide** plays an important antiinflammatory and antiviral role in colds.⁷⁴ Community-acquired, as well as ventilator-associated, sinusitis decreases exhaled nasal nitric oxide^{19,75,76} by reducing its antral production (see Fig. 47-1). Sepsis originating in other organs also markedly inhibits the antral production of **nitric oxide** through downregulation of inducible **nitric oxide** synthetase messenger ribonucleic acid.¹⁹ The sepsis-induced reduction of antral **nitric oxide** concentrations contributes to impaired mucociliary clearance and decreased perfusion of the maxillary epithelium, both factors that facilitate infection of the maxillary cavity by impairing bacterial cleansing.⁷⁷

Epidemiology and Complications

Incidence

In critically ill patients, the reported incidence of radiographic sinusitis varies from 25% to 75%.⁷⁸ More than 80% of radiographic abnormalities of the maxillary sinuses are associated with radiographic abnormalities of the ethmoid, sphenoid, and frontal sinuses.^{8,79} Variability in the estimated incidence stems from the many radiographic techniques used for diagnosis: Conventional radiography is much less accurate than computed tomography or maxillary ultrasound. After 12 hours of nasal endotracheal intubation and/or nasogastric tube placement, 38% of critically ill patients have computed tomographic evidence of radiographic maxillary sinusitis.⁸ Among factors other than the presence of a foreign body in the nostrils, the supine position, head⁶³ and/or facial trauma,^{51,53,80} allergy, and sepsis have been incriminated.⁸⁰ After 7 days, the incidence of radiographic maxillary sinusitis increases to 80%.⁸ Infectious sinusitis is less frequent than radiographic sinusitis, occurring only in 20% to 30% of patients intubated longer than 7 days.⁷⁹ Its incidence is higher in nasotracheally than in orotracheally intubated patients.⁸

Nosocomial Sinusitis and Ventilator-Associated Pneumonia

A link between infectious maxillary sinusitis and bronchopneumonia has been established experimentally.^{47,82} In critically ill patients, nosocomial sinusitis is considered a major cause of ventilator-associated pneumonia.^{8,13,56,62} In a randomized study assessing a systematic search for sinusitis in nasotracheally ventilated patients, the incidence of ventilator-associated pneumonia was significantly higher in the control group than in the study group, where infectious maxillary sinusitis was systematically sought and treated, when confirmed, by sinus drainage and intravenous antibiotics.¹¹ In addition, the mortality rate significantly decreased from 46% in the control group to 36% in the study group. Although the association between nosocomial sinusitis and ventilator-associated pneumonia appears highly likely, the frequent discordance between microbiologic results from sinus and lower airway cultures^{8,11} suggests that these two infections may also arise simultaneously and independently because of shared risk factors and diminished host defense for infection. A heavily contaminated nasopharynx may be the common bacterial reservoir from which sinuses and lungs are infected. Conversely, antral infection, characterized by high concentrations of microorganisms,⁸ can also be a reservoir from which the oropharynx, the tracheobronchial tree, and the lung parenchyma are secondarily infected. Interestingly, concentrations of **nitric oxide** in the maxillary and nasal cavities markedly increase as infectious maxillary sinusitis is treated with antibiotics and sinus lavage.⁸³ The restoration of normal local concentrations of **nitric oxide**, by reestablishing maxillary ciliary function and immune defense, may help in preventing maxillary cavity superinfection and eliminating one of the risk factors for ventilator-associated pneumonia.

Diagnosis

Clinical Diagnosis

Unlike clinical symptomatology of community-acquired maxillary sinusitis, clinical signs of ventilator-associated maxillary sinusitis are scarce and of limited specificity.⁷⁸ General signs of infection such as fever and leukocytosis have poor specificity. Mucopurulent nasal discharge is evocative but of limited sensitivity.⁵¹ Frontal headache is often blunted by sedative drugs aimed at facilitating mechanical ventilation. As a consequence, ventilator-associated maxillary sinusitis cannot be reliably detected on clinical signs alone and should be systematically sought in the presence of fever of unknown origin.^{4,11,84}

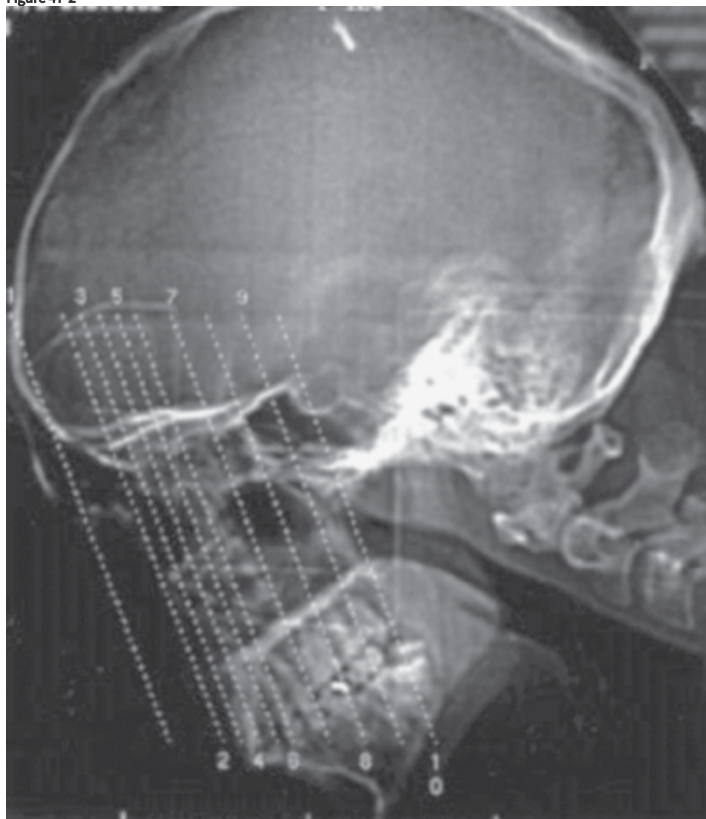
Radiographic Diagnosis

Rather than standard sinus radiography using the Blondeau's view performed in the upright position, sinus ultrasound is the first diagnostic tool for demonstrating community-acquired acute maxillary sinusitis in ambulatory and spontaneously breathing patients, and ventilator-associated maxillary sinusitis in ventilated patients lying in the supine position.^{85,86}

Computed Tomography Scan

Paranasal computed tomography is the reference technique for establishing the diagnosis of radiographic sinusitis.^{8,87-91} A low-dose computed tomography examination is preferred to avoid excessive radiation exposure.⁹¹ Ten thin, noncontiguous computed tomography sections, using an initial interslice gap of 15 mm aimed at avoiding direct radiation to the eye lens (Fig. 47-2), should be performed. Such a technique decreases radiation exposure by more than 90% without seriously affecting the quality of the images (Fig. 47-3).

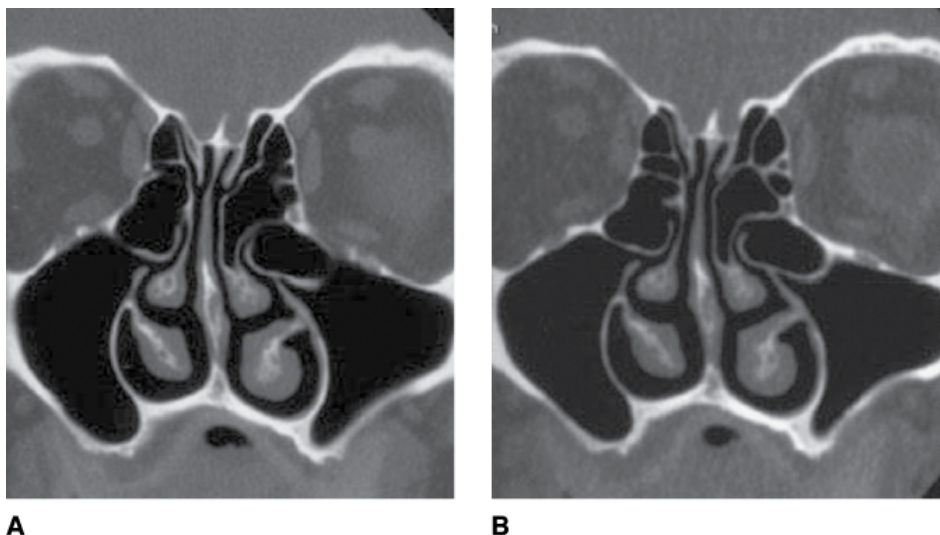
Figure 47-2



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Scout view with different interslice gap between the 10 slices of the low-dose computed tomography protocol recommended for diagnosis of radiographic sinusitis. (Reproduced, with permission, from Hagtvedt T, Aalokken TM, Notthellen J, Kolbenstvedt A. A new low-dose CT examination compared with standard-dose CT in the diagnosis of acute sinusitis. *Eur Radiol*. 2003;13:976-980. With kind permission from Springer Science and Business Media.⁹¹)

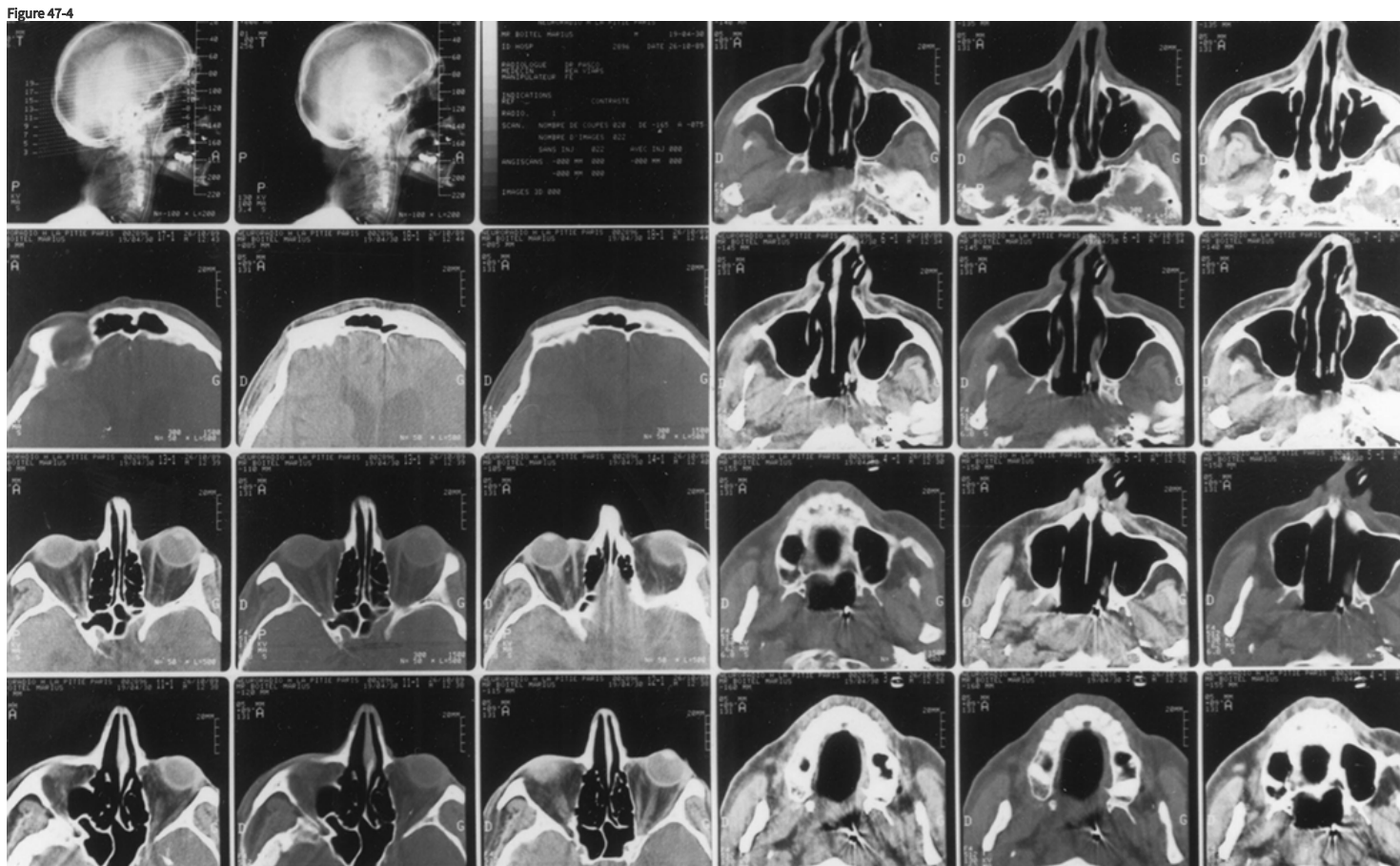
Figure 47-3



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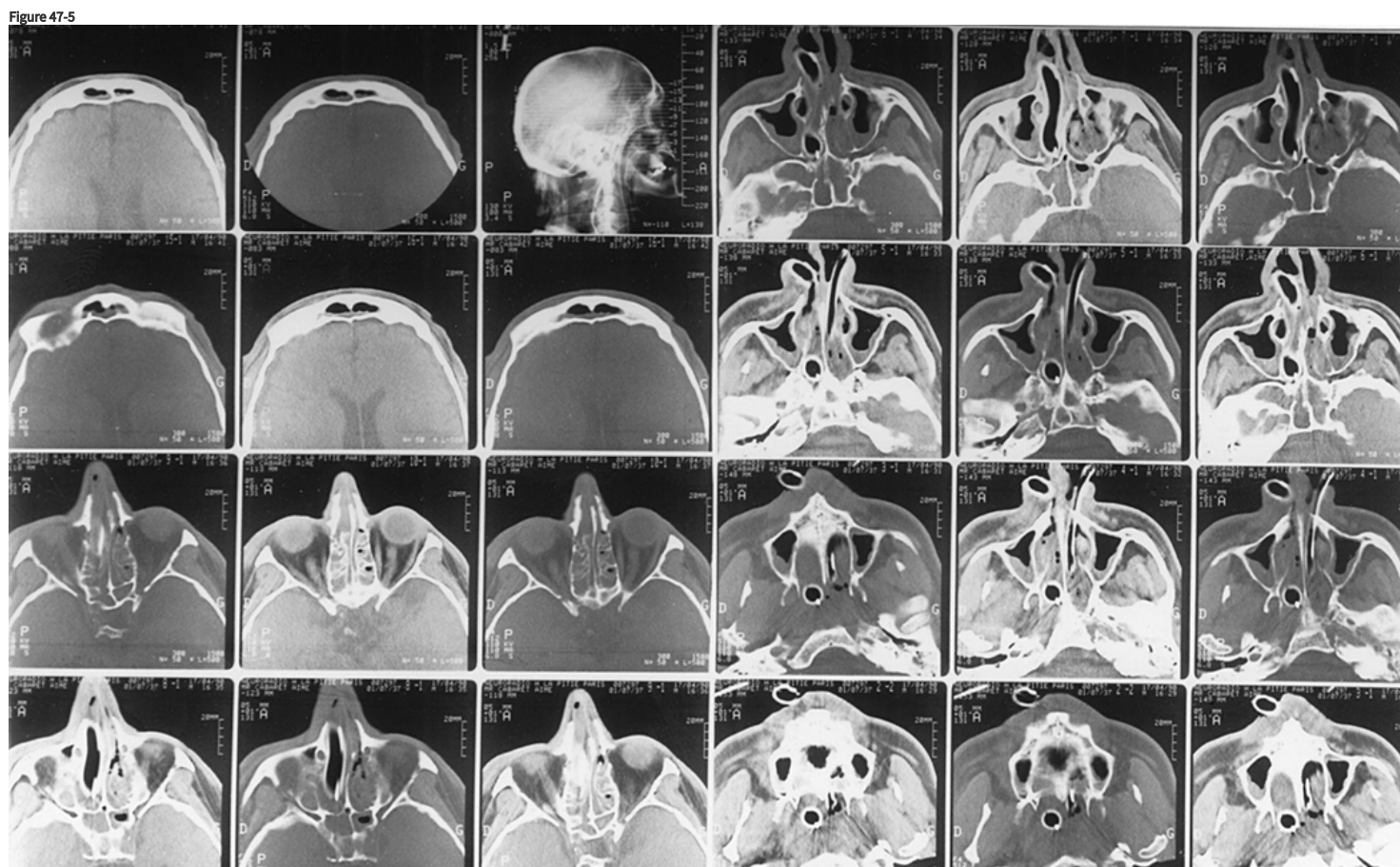
Standard-dose. (A) compared to low-dose (B) computed tomography images of normal ethmoid and maxillary sinuses obtained in a coronal plane through the ostiomeatal complex. (Reproduced, with permission, from Hagtvedt T, Aalokken TM, Notthellen J, et al. A new low-dose CT examination compared with standard-dose CT in the diagnosis of acute sinusitis. *Eur Radiol*. 2003;13:976–980. With kind permission from Springer Science and Business Media.)

In ventilated patients, the computed tomographic image of a given paranasal sinus can be classified into three categories:⁸ normal (Fig. 47-4), mucosal thickening (Fig. 47-5), and radiographic maxillary sinusitis, characterized by a liquid content within the sinus cavity (Fig. 47-6). More than 80% of patients with radiographic abnormalities of maxillary sinuses have also abnormal frontal, ethmoid, and sphenoid sinuses. Conversely, only half of ventilated patients with normally aerated maxillary sinuses have radiographic abnormalities of frontal, ethmoid, and sphenoid sinuses.⁸ An air–fluid level in the sinus cavity or the complete sinus opacification are evocative of infectious sinusitis and call for a bacteriologic confirmation. The significance of mucosal thickening in ventilated patients is incompletely understood. Very frequently, it represents an early stage of radiographic sinusitis.⁸



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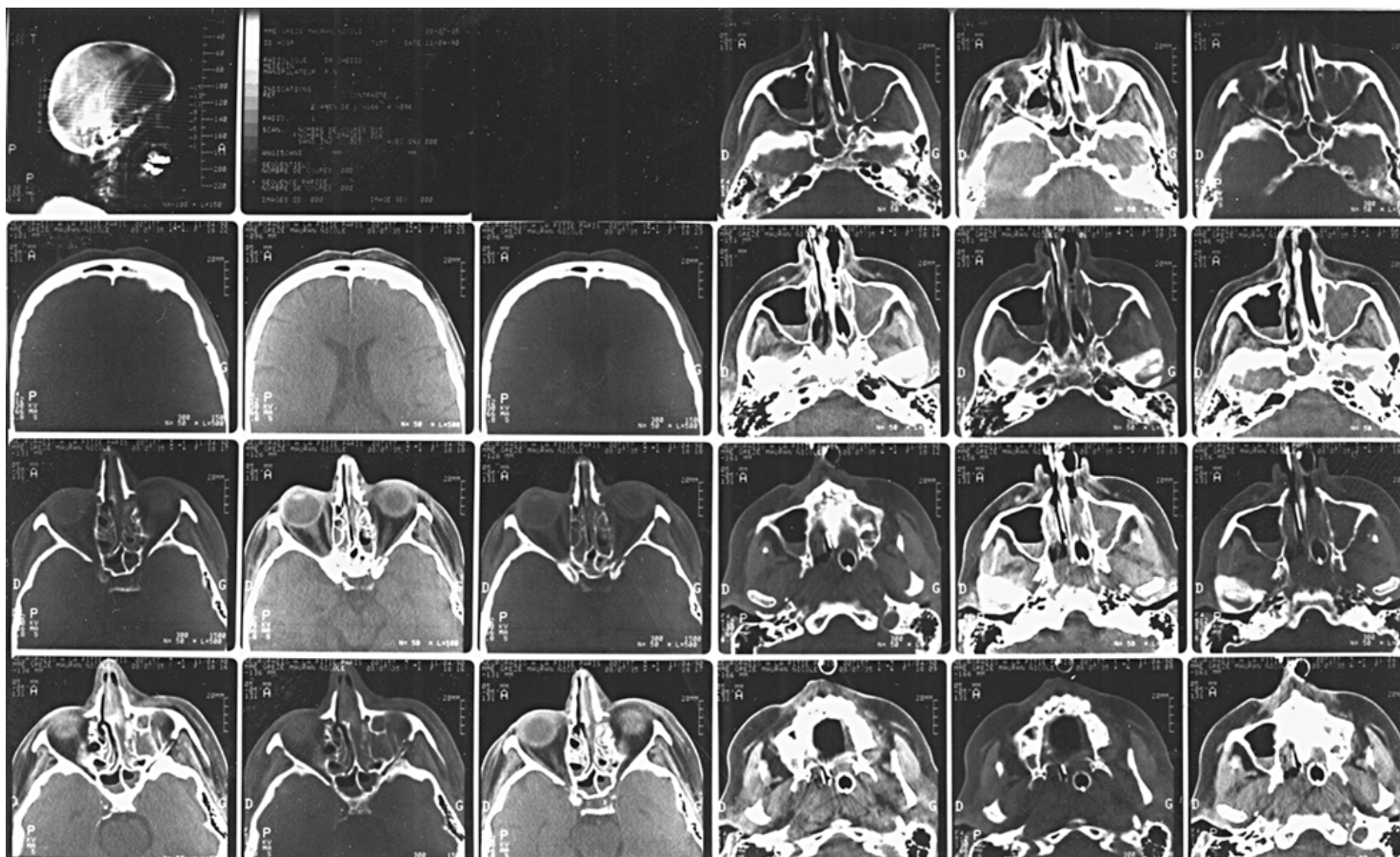
Normally aerated maxillary, ethmoid, sphenoid, and frontal sinuses in a ventilated patient with orotracheal intubation and the gastric tube positioned in the left nostril. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Rouby JJ, Laurent P, Gosnach M, et al. Risk factors and clinical relevance of nosocomial maxillary sinusitis in the critically ill. *Am J Respir Crit Care Med*. 1994;150:776–783. Official Journal of the American Thoracic Society.)



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Computed tomography scan showing mucosal thickening of the right and left maxillary and frontal sinuses, and of the left ethmoid and sphenoid sinuses, associated with complete opacification of the right ethmoid and sphenoid sinuses in a ventilated patient with right nasotracheal intubation and a gastric tube positioned in the left nostril. On a second paranasal computed tomography scan, performed 1 week later without changing the initial position of the endotracheal and gastric tubes, complete opacification of the maxillary, ethmoid, and sphenoid sinuses was observed with a persistent mucosal thickening of both frontal sinuses. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Rouby JJ, Laurent P, Gosnach M, et al. Risk factors and clinical relevance of nosocomial maxillary sinusitis in the critically ill. 1994;150:776–783. Official Journal of the American Thoracic Society.)

Figure 47-6



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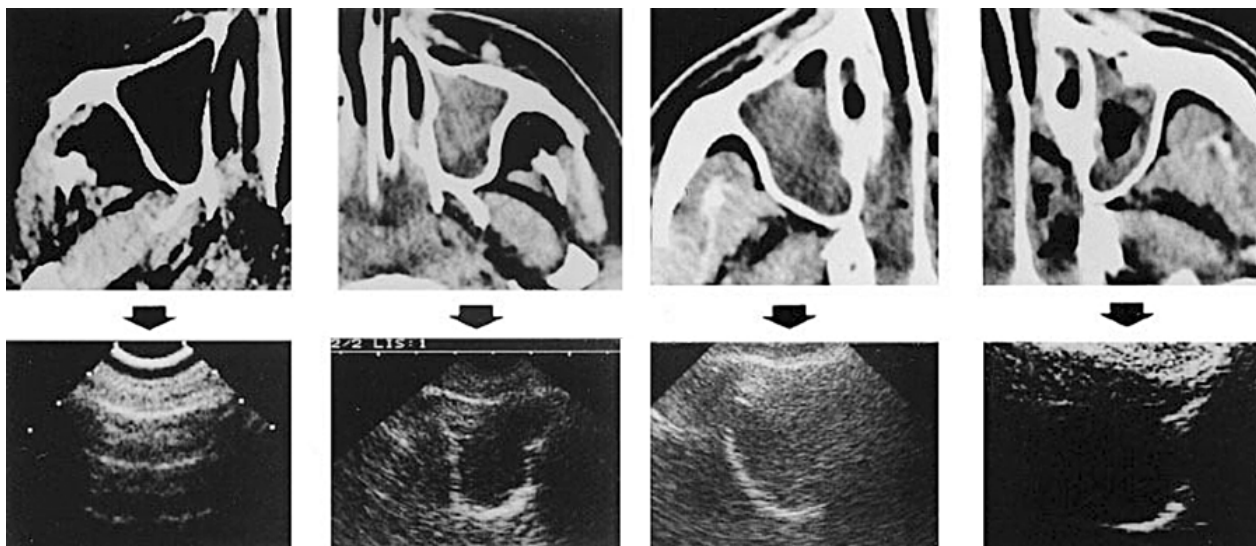
Typical aspect of radiographic maxillary sinusitis in a ventilated patient with left nasotracheal intubation and a right nasogastric tube. The left maxillary sinus exhibits complete opacification, whereas the right maxillary sinus exhibits a characteristic air-fluid level. Mucosal thickening can be seen in both ethmoid sinuses and the left frontal sinus. Both sphenoid sinuses exhibit air-fluid levels, whereas the right frontal sinus is normally aerated. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Rouby JJ, Laurent P, Gosnach M, et al. Risk factors and clinical relevance of nosocomial maxillary sinusitis in the critically ill. *Am J Respir Crit Care Med*. 1994;150:776–783. Official Journal of the American Thoracic Society.)

Although considered the gold standard, several limitations preclude the routine use of computed tomography for diagnosing ventilator-associated sinusitis. It requires transportation of ventilated patients outside the intensive care unit, a risky procedure.⁹² It is costly and not easily repeatable. As a consequence, ultrasound techniques have been developed over the last 15 years and are currently considered an attractive alternative diagnostic tool.^{61,88–90,93}

Ultrasound Examination of Maxillary Sinuses

A-mode sinus ultrasound was proposed in the early 1990s for diagnosing radiographic maxillary sinusitis in ventilated patients.⁵¹ At present, bedside B-mode ultrasound is considered as the reference technique.^{88,89,93,94} The patient should be examined at the bedside in a semirecumbent position with a 3.5-MHz probe. The ultrasonic procedure should be performed in a transversal plane using different probe angulations, at the level of the front maxillary sinus wall, delineated by the lower orbital border, the nose, the upper maxilla, and the external cheekbone.⁸⁹ Image formation depends on the presence of fluid or mucosal thickening within the sinus cavity that transmits the ultrasonic beam to deep anatomic structures. When the posterior bony wall of the maxillary sinus becomes visible in part or completely, depending on the amount of gas present in the sinus cavity, a “sinusogram” is observed (Fig. 47-7).

Figure 47-7

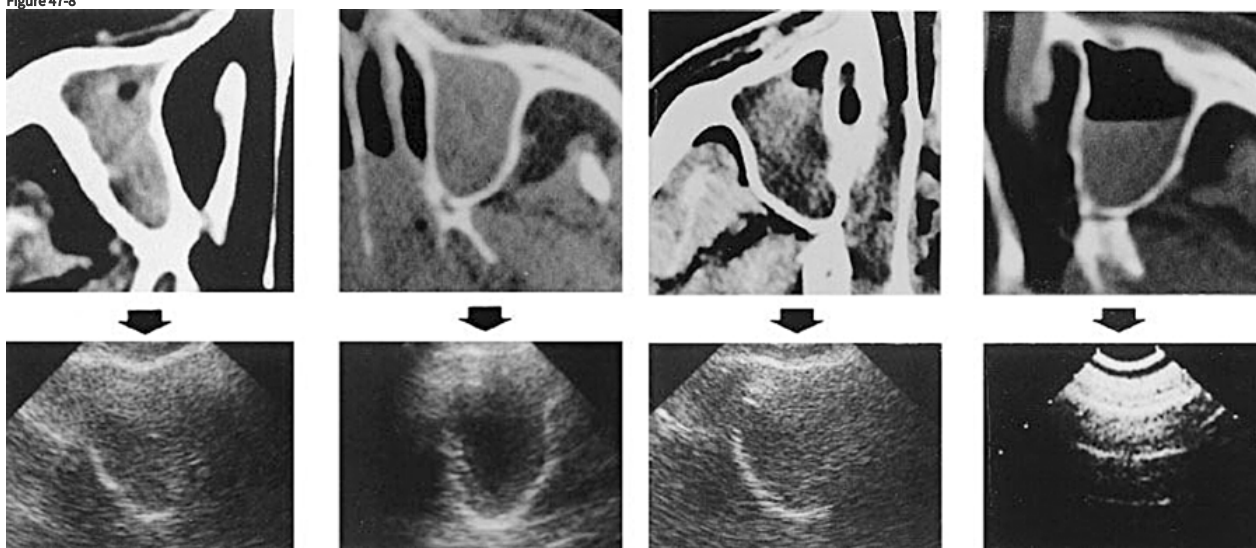


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Normal versus pathologic ultrasound pattern. Computed tomography sections of maxillary sinuses are in the upper row with their corresponding ultrasound pattern in the lower row. From left to right: normal pattern—total opacity of the left maxillary sinus (the internal, posterior, and external walls of the sinus are visible, giving a complete “sinusogram”); 90% air–fluid level of the right maxillary sinus (the internal wall is not visualized, giving an incomplete “sinusogram”); and mucosal thickening of the left maxillary sinus (the internal wall is not visible and the external wall is ill-defined, giving a partial “sinusogram”). (Reproduced, with permission, from Lichtenstein D, Biderman P, Meziere G, et al. The “sinuso-gram”, a real-time ultrasound sign of maxillary sinusitis. *Intensive Care Med.* 1998;24:1057–1061. With kind permission from Springer Science and Business Media.)

In ventilated patients, the ultrasound image of the maxillary sinus can be classified into three categories:^{89,93,94} absence of “sinusogram”, partial “sinusogram”, and complete “sinusogram” (Fig. 47-8). As shown in Figures 47-7 and 47-8, the absence of a “sinusogram” in a ventilated patient lying in the supine position may correspond either to an air–fluid level or to a normally aerated sinus. Repeating the ultrasound procedure in the half-sitting position, however, may reveal an incomplete “sinusogram”, thereby asserting the diagnosis of radiographic maxillary sinusitis.^{90,93} A partial “sinusogram” may indicate either mucosal thickening, simple inflammation of the sinus mucosa, or an air–fluid level. Note that in half the patients with complete opacification of the maxillary cavity on computed tomography exhibit a partial “sinusogram”, resulting from the presence of small bubbles within the collection.⁸⁹ Repeating ultrasound in the supine position may facilitate the differentiation between mucosal thickening and true maxillary sinusitis.⁹³ The disappearance of the partial “sinusogram” in the supine position is evocative of an air–fluid level, characteristic of radiographic maxillary sinusitis. In contrast, the persistence of a partial “sinusogram” in the supine position is strongly suggestive of mucosal thickening.

Figure 47-8



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Different abnormal ultrasound patterns. Computed tomography sections of maxillary sinuses are in the upper row with their corresponding ultrasound pattern in the lower row. From the left to the right: total opacity of the right maxillary sinus with a small anterior air bubble (the external wall is visible and the internal wall is not visible, giving an incomplete “sinusogram”); total opacity of the left maxillary sinus (the internal, posterior, and external walls of the sinus are visible, giving a complete “sinusogram”); 90% air–fluid level of the right maxillary sinus (the internal wall is not visualized, giving an incomplete

“sinusogram”); and 50% air–fluid level of the left maxillary sinus (the walls are not visible, giving an acoustic barrier, and thus no “sinusogram”). (Reproduced, with permission, from Lichtenstein D, Biderman P, Meziere G, et al. The “sinuso-gram”, a real-time ultrasound sign of maxillary sinusitis. *Intensive Care Med.* 1998;24:1057–1061. With kind permission from Springer Science and Business Media.)

If radiographic doubt persists and clinical signs are evocative of infectious maxillary sinusitis, a paranasal computed tomography scan should be performed. A complete “sinusogram” is always related to a true radiographic maxillary sinusitis and requires antral puncture in the presence of fever of unknown origin. Based on these recommendations, bedside maxillary ultrasound has a sensitivity of 100% and a specificity of 97% for diagnosing radiographic maxillary sinusitis in intubated patients undergoing mechanical ventilation.⁹⁰ An important limitation of bedside maxillary ultrasound is that it does not provide any reliable information on frontal, ethmoid, and sphenoid sinuses.

Microbiologic Diagnosis

Diagnostic Criteria for Infectious Maxillary Sinusitis

Infectious maxillary sinusitis has to be clearly differentiated from radiographic maxillary sinusitis. The latter corresponds to fluid accumulation within the antral cavity, the liquid being either mucus or pus. Infectious sinusitis is defined by the presence of pus in the maxillary cavity and requires cellular and bacterial analysis of the sinus contents. Therefore, antral puncture is indispensable for an accurate diagnosis. Examination of the sinus contents is impossible for frontal and ethmoid sinuses. It requires an easy-to-perform antral puncture for the maxillary sinus and a highly specialized endoscopic puncture for the sphenoid sinus. Maxillary puncture can be performed either by the transnasal or the canine fossa routes.

The transnasal and canine fossa approaches are exposed to massive bacterial contamination issued from the nostrils and gingiva, which are heavily contaminated by nosocomial microorganisms in critically ill patients. The presence of a foreign body in the nose further increases the bacterial burden and seriously complicates the process of nasal decontamination.⁸ After a meticulous and time-consuming disinfection of the nasal cavities, only 50% of the nostrils are sterile and 11% remain heavily contaminated.⁸ As a result, transnasal puncture of the maxillary sinus carries a significant risk of introducing bacteria into the antral cavity and overestimating the likelihood of infectious sinusitis.⁸⁰ As a consequence, a rigid protocol should be followed when performing a transnasal puncture and strict diagnostic criteria required for establishing the diagnosis. After a careful and prolonged disinfection of the nasal cavity using a **povidone-iodine** or a chlorhexidine **alcohol** solution,^{8,95} a nasal swab is performed to assess the efficacy of nasal disinfection. Then transnasal puncture is performed using an Albertini trocar placed below the inferior turbinate, and sinus contents are directly aspirated before any lavage. The recovered fluid is then examined, using cell and quantitative bacterial analysis preceded by a Gram stain. In addition, the nasal swab is analyzed using a semiquantitative bacteriologic analysis. The diagnosis of infectious sinusitis requires fulfillment of the following criteria: more than five altered polymorphonuclear leukocytes per oil immersion field and a positive culture with a bacterial concentration equal to or greater than 10^3 colony-forming units (CFU)/mL in the case of negative nasal swab, or equal to or greater than 10^4 CFU/mL in the case of positive nasal swab.⁸

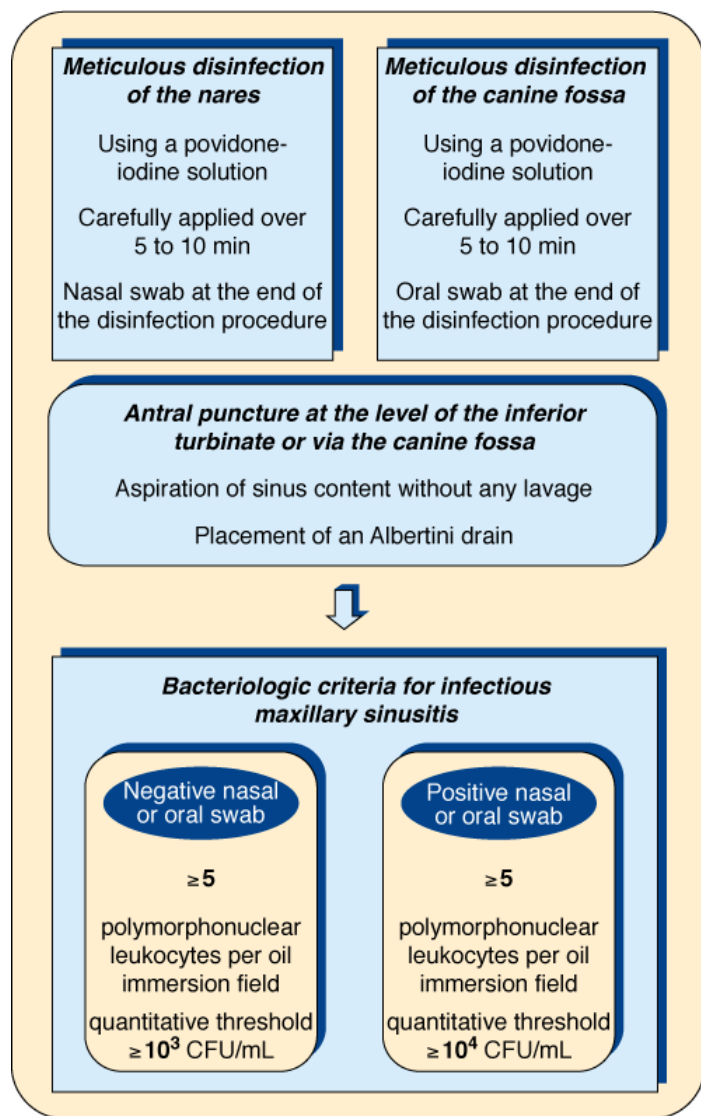
Microbiology of Ventilator-Associated Maxillary Sinusitis

Community-acquired maxillary sinusitis is predominantly caused by *Streptococcus pneumoniae*, *Moraxella catarrhalis*, and *Haemophilus influenzae*. In contrast, the most frequent microorganisms recovered in ventilated adult patients with infectious maxillary sinusitis, are gram-negative bacteria such as Enterobacteriaceae species, gram-positive cocci such as staphylococci or streptococci, anaerobes, and yeasts.^{8,11,81,95–97} Very frequently, infection of the antral cavity is polymicrobial.^{8,62,79,84} Anaerobes are the causative pathogens in more than 60% of cases of infectious maxillary sinusitis, either in association with aerobic species or alone.⁹⁸

In adult patients, the most common anaerobes infecting the antral cavity are *Prevotella* species and *Fusobacterium nucleatum*, all bacteria that belong to the commensal nasal flora and are known to produce β -lactamases, which reduces their sensitivity to antibiotics.⁹⁸ Similar findings have been reported in ventilated children with nosocomial sinusitis: Aerobes were present in only 40% of the patients, anaerobes in only 25% of the patients, and mixed aerobic and anaerobic flora in 35% of the patients.⁹⁷ The predominant aerobes were *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Escherichia coli*, and *Klebsiella pneumoniae*. The predominant anaerobes were *Peptostreptococcus* species, *Prevotella* species, and *Fusobacterium* species.

Anaerobes are more commonly isolated from sinus aspirate samples obtained after prolonged mechanical ventilation.⁹⁷ The high incidence of anaerobes and their underestimation by classic means of bacterial examination probably explain the high number of purulent antral secretions remaining sterile on culture.⁸⁴ The presence of a nasal foreign body (endotracheal or gastric tube) markedly increases bacterial colonization of the ipsilateral middle meatus.⁸⁴ In **Figure 47-9**, a logical approach, including diagnostic criteria, is proposed for establishing the diagnosis of infectious maxillary sinusitis.

Figure 47-9



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Chart summarizing a logical diagnostic approach and proposing criteria for infectious maxillary sinusitis. (Reproduced, with permission, from Lichtenstein D, Biderman P, Meziere G, et al. The "sinuso-gram", a real-time ultrasound sign of maxillary sinusitis. *Intensive Care Med*. 1998;24:1057–1061. With kind permission from Springer Science and Business Media.)

Treatment

Prevention of Ventilator-Associated Sinusitis

Although some of the published data on the prevention of sinus infection in ventilated patients are conflicting,^{8,56,61,62} the following measures can be recommended for reducing the incidence of ventilator-associated sinusitis. First, nasotracheal intubation and nasogastric tubes should be avoided. Most prospective studies that have examined the influence of nasotracheal intubation on ventilator-associated sinusitis have shown an increased incidence of maxillary sinusitis in nasotracheally intubated patients.^{8,56,61} On the other hand, in a randomized single-center study that included 300 ventilated patients, there was no statistically significant difference in the occurrence of ventilator-associated sinusitis between patients intubated via the nasal versus the oral route. A trend ($p = 0.08$), however, suggested less sinusitis in the orotracheal group.⁶² Second, the nostrils should remain free of foreign bodies and be regularly cleaned of nasal secretions that tend to accumulate. In a randomized open-label study performed in 79 patients with multiple trauma, the combination of locally applied nasal α -adrenergic and corticosteroids reduced the incidence of radiologic maxillary sinusitis by 29%.⁹⁹ The topical application of a nasal decongestant, together with locally applied corticosteroids to avoid rhinitis medicamentosa, may reduce ostial obstruction and edema of the ostiomeatal complex that initiates sinus inflammation and mucus accumulation. Further studies are required to confirm the efficacy of this mode of prevention. Third, the patient should be kept preferentially in the semirecumbent position to ensure patency of the different ostia. The observance of these preventive measures does not, however, fully protect against ventilator-associated sinusitis. General sepsis, restrictions in regional blood flow resulting from circulatory shock, as well as allergy and systemic inflammation, all factors encountered in the critical care environment, certainly contribute to antral disease and secondary infection of sinus cavities.

Treatment of Infectious Maxillary Sinusitis

Classically, sinus infection diagnosed in ventilated patients requires the administration of intravenous antibiotics.^{11,62,78} Removal of nasal foreign bodies, semirecumbent positioning, sinus drainage, topical decongestants, and serum lavage of the antral cavity through the drains are useful adjunctive therapies.

Penetration of Antibiotics into Antral Mucosa

Two studies have shown adequate sinus deposition of intravenous antibiotics in patients with ventilator-associated maxillary sinusitis.^{100,101} In a series of six ventilated patients, a single intravenous dose of piperacillin 4 g achieved bactericidal concentrations for up to 8 hours in the sinus fluid obtained by maxillary drainage.¹⁰⁰ In a series of twenty ventilated patients with ventilator-associated sinusitis, a single intravenous dose of amikacin 15 mg/kg achieved high peak concentrations in the sinus fluid obtained by maxillary drainage.¹⁰¹ These encouraging results, however, were challenged by a subsequent study performed in seven critically ill patients with inflammatory and bacterial maxillary sinusitis.¹⁰² In all patients, extracellular tissue concentrations of cefuroxime, ampicillin, and vancomycin measured on biopsy samples obtained by sinuscopy were lower than the corresponding plasma concentrations. In patients with noninfectious maxillary sinusitis, the ratio between tissue and plasma concentrations varied from 10% to 73%. In one patient treated with intravenous cefuroxime for a nasogastric tube-induced purulent maxillary sinusitis, very low plasma concentrations with no measurable mucosal concentrations were found. In one patient treated with intravenous ampicillin for a nasotracheal tube-induced purulent maxillary sinusitis, high tissue concentrations were found, reaching 95% of the plasma concentration in the left antral mucosa and 53% in the right antral mucosa. In one tracheostomized patient treated with vancomycin for a left purulent maxillary sinusitis, high mucosal concentrations of vancomycin were found. Surprisingly, in the three patients with purulent maxillary sinusitis, bacterial cultures of the sinus biopsies were positive, irrespective of the antibiotic tissue concentrations.

Sinus Drainage and Antibiotic Therapy

In critically ill patients, infectious maxillary sinusitis frequently emerges despite prior antibiotic administration to which the causative microorganisms are sensitive,^{95,102} suggesting that antibiotics may not be a sufficient therapy for ventilator-associated maxillary sinusitis. Several hypotheses have been proposed for explaining the persistence of bacteria within the antral cavity despite the intravenous administration of appropriate antibiotics. As demonstrated in a small number of patients, antral penetration of systemic antibiotics is very variable and mucosal concentrations may remain below minimal inhibitory concentrations.¹⁰² In patients on long-term mechanical ventilation, the presence of a chronic anaerobic infection might be associated with biofilm formation protecting the bacteria against the bactericidal effects of antibiotics.¹⁰² Measurements of antral antibiotic concentrations require sinuscopy and/or the insertion of drains into the sinus cavity.^{100–102} Such invasive procedures may create a regional inflammatory reaction responsible for increased antral antibiotic concentrations that may no longer be representative of true tissue concentrations obtained in the intact sinus cavity.

The logical implication of these clinical findings and hypotheses is that sinus drainage should be adopted as a first-line therapy for ventilator-associated maxillary sinusitis, together with daily sinus lavage and topical antibiotic administration.^{8,95} To avoid recurrence of infectious maxillary sinusitis, sinus drains should be left in place until endotracheal extubation and definitive weaning from mechanical ventilation.⁸ Removing foreign bodies from the nostrils and positioning the patient in the semirecumbent position may help recovery from antral disease.

Important Unknowns

Clinical Relevance of Frontal, Ethmoid, and Sphenoid Sinusitis

Radiographic abnormalities of the ethmoid and sphenoid sinuses are found in more than 80% of ventilated patients.⁸ In ventilated patients with computed tomography-delineated abnormalities of maxillary sinuses, more than 95% have radiographic ethmoid and sphenoid sinusitis. Half of ventilated patients with normal maxillary sinuses have computed tomography-delineated abnormalities (mucosal thickening or complete opacification of the sinus cavities) of the ethmoid and sphenoid sinuses.⁸ The meaning and clinical impact of these abnormalities remain unknown. Despite sinus drainage associated with topical administration of appropriate antibiotics and the absence of other sites of infection, more than 10% of ventilated patients with documented infectious maxillary sinusitis remain febrile with persistent signs of sepsis, raising the possibility of a residual infection within the ethmoid and sphenoid cavities.⁸ Although community-acquired infectious sphenoid sinusitis remains a rare entity,^{103,104} further studies are required for assessing the exact incidence of infectious ethmoid and sphenoid sinusitis as a cause of occult fever in ventilated patients.

Role of Antibiotics in the Therapy of Ventilator-Associated Maxillary Sinusitis

There is an ongoing controversy as to whether intravenous or topical antibiotics should be administered to ventilated patients with documented infectious maxillary sinusitis. Available data on the penetration of systemic antibiotics into the diseased antral mucosa of ventilated patients are scarce and are based on very few critically ill patients.^{100–102} In addition, the role of viruses in the infection of sinus cavities remains unknown. As with community-acquired acute maxillary sinusitis, the clinical course of which is not influenced by antibiotic treatment,¹⁰⁵ ventilator-associated infectious maxillary sinusitis might be caused by viruses and require specific antiviral therapy rather than intravenous antibiotics. Clinical data support the concept that radiographic maxillary

sinusitis is the first step along the path to an established sinus infection⁸ and raise the issue of whether removal of nasal foreign bodies and placement of sinus drainage catheters should not be undertaken before frank infection and sepsis occur.¹⁰ Further studies are required to assess factors influencing the penetration of antibacterial agents into the sinus cavities and to clarify many uncertainties concerning treatment optimization of ventilator-associated sinusitis.

Conclusions and Future Prospects

Infection of paranasal sinuses is an important cause of occult sepsis in critically ill patients receiving mechanical ventilation. It is a typical nosocomial infection resulting mainly from the presence of nasal foreign bodies. Mechanical and edematous obstruction of the sinus ostia, suppression of the normal sinus ventilation, inhibition of the maxillary production of [nitric oxide](#), and cessation of the normal sinus mucociliary clearance are different factors causing an inflammatory antral disease that becomes rapidly superinfected by pathogens proliferating around nasotracheal and nasogastric tubes. Like any nosocomial infection that may weigh heavily against patients' chances of survival, an active policy of detection and prevention should be undertaken and not lead to antibiotic overuse. Faced with an unknown cause of sepsis, the clinician should systematically perform maxillary sinus ultrasonography at the bedside, searching for radiographic maxillary sinusitis. When present, maxillary sinus puncture should be performed with careful attention to avoid nasal contamination of sinus contents. Once confirmed by quantitative bacteriology and cell analysis, infectious maxillary sinusitis should be treated by prolonged sinus drainage, removal of nasal foreign bodies, positioning in the semirecumbent position, and administration of intravenous or topical antibiotics. If applied early after initiation of mechanical ventilation, orotracheal intubation, oral placement of gastric tubes, and semirecumbent positioning may decrease the incidence of ventilator-associated maxillary sinusitis. A large multicenter randomized study should be undertaken to assess whether these preventive measures are associated with a decrease in the incidence of ventilator-associated pneumonia and mortality.

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